

Answers

1.A	2.B	3.B	4.6	5.5	6.A	7.6	8.B
9.B	10.C	11.C	12.D	13.D	14.A	15.D	16.D
17.2	18.1	19.2	20.6	21.C	22.A	23.33	24.C

Explanations

1. A

Given that each item is sold for a profit of 2 times the number of materials required for the dish.

Hence for different finished products: The cost price and selling price are :

Finished Product	Cost Price	Profit	Selling Price
Mango Smoothie	8	4	12
Apple Smoothie	7	4	11
Banana Smoothie	6	4	10
Mixed Fruit	11	8	19
Fruit Salad	6	6	12

Given that in any hour only either zero or one unit of a single finished product is sold. Hence the price distribution for the first three hours sales distribution is given by :

Business Hour	Ganga	Kaveri	Narmada
Hour 1	11 + 12	19	19+12
Hour 2	11+ 10	12+ 10	11 + 10
Hour 3	19+ 10	19+ 11	11 + 12

Additionally, it has been mentioned that no one except Ganga possibly sold Mango smoothies.

Considering Apple Smoothie : (AS), Mango Smoothie : (M.S), Banana Smoothie : (B. S), Mixed Fruit : (M.F), Fruit Salad : (F.S)

Business Hour	Ganga	Kaveri	Narmada
Hour 1	AS + F.S/M.S	M.F	M.F + F.S
Hour 2	AS + B.S	F.S + B.S	AS + B.S
Hour 3	M.F+ B.S	M.F + AS	AS + F.S

In the last one hour everyone had one unit each of two different materials as leftovers:

Each leftover item is sold at Re1 less than their cost price. Hence

One unit of milk, mango, and apple will cost ₹4, ₹2, ₹1,

The three possible combinations for the last hour in which raw materials can be sold is :

One unit of milk+ One unit of mango = Rs 4+2 = Rs 6.

One unit of milk+ One unit of apple = Rs 4+1 = Rs 5.

One unit of mango+ One unit of apple = Rs 2+1 = Rs 3.

In the last one hour :

Total sale of Ganga is Rs 30. This is the total cost earned by considering selling the finished items and raw materials :

Hence the possible cases are :

(24 + 6) or (25 + 5) or (27 + 3). It is not possible to earn 25 and 27 by selling finished products.

Hence she earns Rs 24 which is possible by selling one (M.S + F.S) and one raw unit of milk and mango each.

Total sale of Kaveri is Rs 27. This is the total cost earned by considering selling the finished items and raw materials :

Hence the possible cases are :

(21 + 6) or (22+5) or (24+3).

Since Kaveri cannot sell mango smoothies the possible cases are :

(21+6) Selling one apple smoothie and one banana smoothie and one raw unit of Milk and Mango.

(22+5) Selling one Banana smoothie and Fruit Salad, one unit of raw milk, and one unit of Raw Apple.

Total sale for Narmada is Rs 22. The possible cases are :

(16+6), (17+5), (19+3).

Among these, the only possible case is 19 + 3. Selling one unit Mixed Fruit, and one raw unit of Mango and Apple.

Case 1 :

Business Hour	Ganga	Kaveri	Narmada
Hour 1	A.S + F.S/M.S	M.F	M.F + F.S
Hour 2	A.S + B.S	F.S + B.S	A.S + B.S
Hour 3	M.F+ B.S	M.F + A.S	A.S + F.S
Final Hour	M.S+F.S+Milk+Mango	A.S+B.S+Milk+Mango	M.F+Mango+ Apple

case 2 :

Business Hour	Ganga	Kaveri	Narmada
Hour 1	A.S + F.S/M.S	M.F	M.F + F.S
Hour 2	A.S + B.S	F.S + B.S	A.S + B.S
Hour 3	M.F+ B.S	M.F + A.S	A.S + F.S
Final Hour	M.S+F.S+Milk+Mango	F.S+B.S+Milk+Apple	M.F+Mango+ Apple

In the first one hour, either one or two fruit salads are sold.

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2. B

Given that each item is sold for a profit of 2 times the number of materials required for the dish.

Hence for different finished products: The cost price and selling price are :

Finished Product	Cost Price	Profit	Selling Price
Mango Smoothie	8	4	12
Apple Smoothie	7	4	11
Banana Smoothie	6	4	10
Mixed Fruit	11	8	19
Fruit Salad	6	6	12

Given that in any hour only either zero or one unit of a single finished product is sold. Hence the price distribution for the first three hours sales distribution is given by :

Business Hour	Ganga	Kaveri	Narmada
Hour 1	11 + 12	19	19+12
Hour 2	11+ 10	12+ 10	11 + 10
Hour 3	19+ 10	19+ 11	11 + 12

Additionally, it has been mentioned that no one except Ganga possibly sold Mango smoothies.

Considering Apple Smoothie : (AS), Mango Smoothie : (M.S), Banana Smoothie : (B. S), Mixed Fruit : (M.F), Fruit Salad : (F.S)

Business Hour	Ganga	Kaveri	Narmada
Hour 1	AS + F.S/M.S	M.F	M.F + F.S
Hour 2	AS + B.S	F.S + B.S	AS + B.S
Hour 3	M.F+ B.S	M.F + AS	AS + F.S

In the last one hour everyone had one unit each of two different materials as leftovers:

Each leftover item is sold at Re1 less than their cost price. Hence

One unit of milk, mango, and apple will cost ₹4, ₹2, ₹1,

The three possible combinations for the last hour in which raw materials can be sold is :

One unit of milk+ One unit of mango = Rs 4+2 = Rs 6.

One unit of milk+ One unit of apple = Rs 4+1 = Rs 5.

One unit of mango+ One unit of apple = Rs 2+1 = Rs 3.

In the last one hour :

Total sale of Ganga is Rs 30. This is the total cost earned by considering selling the finished items and raw materials :

Hence the possible cases are :

(24 + 6) or (25 + 5) or (27 + 3). It is not possible to earn 25 and 27 by selling finished products.

Hence she earns Rs 24 which is possible by selling one (M.S + F.S) and one raw unit of milk and mango each.

Total sale of Kaveri is Rs 27. This is the total cost earned by considering selling the finished items and raw materials :

Hence the possible cases are :

(21 + 6) or (22+5) or (24+3).

Since Kaveri cannot sell mango smoothies the possible cases are :

(21+6) Selling one apple smoothie and one banana smoothie and one raw unit of Milk and Mango.

(22+5) Selling one Banana smoothie and Fruit Salad, one unit of raw milk, and one unit of Raw Apple.

Total sale for Narmada is Rs 22. The possible cases are :

(16+6), (17+5), (19+3).

Among these, the only possible case is 19 + 3. Selling one unit Mixed Fruit, and one raw unit of Mango and Apple.

Case 1 :

Business Hour	Ganga	Kaveri	Narmada
Hour 1	A.S + F.S/M.S	M.F	M.F + F.S
Hour 2	A.S + B.S	F.S + B.S	A.S + B.S
Hour 3	M.F+ B.S	M.F + A.S	A.S + F.S
Final Hour	M.S+F.S+Milk+Mango	A.S+B.S+Milk+Mango	M.F+Mango+ Apple

case 2 :

Business Hour	Ganga	Kaveri	Narmada
Hour 1	A.S + F.S/M.S	M.F	M.F + F.S
Hour 2	A.S + B.S	F.S + B.S	A.S + B.S
Hour 3	M.F+ B.S	M.F + A.S	A.S + F.S
Final Hour	M.S+F.S+Milk+Mango	F.S+B.S+Milk+Apple	M.F+Mango+ Apple

In both, cases Ganga did not sell any leftover apples. The other options are not necessarily true.

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3. B

Given that each item is sold for a profit of 2 times the number of materials required for the dish.

Hence for different finished products: The cost price and selling price are :

Finished Product	Cost Price	Profit	Selling Price
Mango Smoothie	8	4	12
Apple Smoothie	7	4	11
Banana Smoothie	6	4	10
Mixed Fruit	11	8	19
Fruit Salad	6	6	12

Given that in any hour only either zero or one unit of a single finished product is sold. Hence the price distribution for the first three hours sales distribution is given by :

Business Hour	Ganga	Kaveri	Narmada
Hour 1	11 + 12	19	19+12
Hour 2	11+ 10	12+ 10	11 + 10
Hour 3	19+ 10	19+ 11	11 + 12

Additionally, it has been mentioned that no one except Ganga possibly sold Mango smoothies.

Considering Apple Smoothie : (A.S), Mango Smoothie : (M.S), Banana Smoothie : (B. S), Mixed Fruit : (M.F), Fruit Salad : (F.S)

Business Hour	Ganga	Kaveri	Narmada
Hour 1	A.S + F.S/M.S	M.F	M.F + F.S
Hour 2	A.S + B.S	F.S + B.S	A.S + B.S
Hour 3	M.F+ B.S	M.F + A.S	A.S + F.S

In the last one hour everyone had one unit each of two different materials as leftovers:

Each leftover item is sold at Re1 less than their cost price. Hence

One unit of milk, mango, and apple will cost ₹4, ₹2, ₹1,

The three possible combinations for the last hour in which raw materials can be sold is :

One unit of milk+ One unit of mango = Rs 4+2 = Rs 6.

One unit of milk+ One unit of apple = Rs 4+1 = Rs 5.

One unit of mango+ One unit of apple = Rs 2+1 = Rs 3.

In the last one hour :

Total sale of Ganga is Rs 30. This is the total cost earned by considering selling the finished items and raw materials :

Hence the possible cases are :

(24 + 6) or (25 + 5) or (27 + 3). It is not possible to earn 25 and 27 by selling finished products.

Hence she earns Rs 24 which is possible by selling one (M.S + F.S) and one raw unit of milk and mango each.

Total sale of Kaveri is Rs 27. This is the total cost earned by considering selling the finished items and raw materials :

Hence the possible cases are :

(21 + 6) or (22+5) or (24+3).

Since Kaveri cannot sell mango smoothies the possible cases are :

(21+6) Selling one apple smoothie and one banana smoothie and one raw unit of Milk and Mango.

(22+5) Selling one Banana smoothie and Fruit Salad, one unit of raw milk, and one unit of Raw Apple.

Total sale for Narmada is Rs 22. The possible cases are :

(16+6), (17+5), (19+3).

Among these, the only possible case is 19 + 3. Selling one unit Mixed Fruit, and one raw unit of Mango and Apple.

Case 1 :

Business Hour	Ganga	Kaveri	Narmada
Hour 1	A.S + F.S/M.S	M.F	M.F + F.S
Hour 2	A.S + B.S	F.S + B.S	A.S + B.S
Hour 3	M.F+ B.S	M.F + A.S	A.S + F.S
Final Hour	M.S+F.S+Milk+Mango	A.S+B.S+Milk+Mango	M.F+Mango+ Apple

case 2 :

Business Hour	Ganga	Kaveri	Narmada
Hour 1	A.S + F.S/M.S	M.F	M.F + F.S
Hour 2	A.S + B.S	F.S + B.S	A.S + B.S
Hour 3	M.F+ B.S	M.F + A.S	A.S + F.S
Final Hour	M.S+F.S+Milk+Mango	F.S+B.S+Milk+Apple	M.F+Mango+ Apple

In except fruit salad, all the other finished products require one unit of milk :

For case 1: a minimum of 19 units of milk and a maximum of 20 units of milk can be used.

For case 2: a minimum of 18 units of milk and a maximum of 19 units of milk is used

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4.6

Given that each item is sold for a profit of 2 times the number of materials required for the dish.

Hence for different finished products: The cost price and selling price are :

Finished Product	Cost Price	Profit	Selling Price
Mango Smoothie	8	4	12
Apple Smoothie	7	4	11
Banana Smoothie	6	4	10
Mixed Fruit	11	8	19
Fruit Salad	6	6	12

Given that in any hour only either zero or one unit of a single finished product is sold. Hence the price distribution for the first three hours sales distribution is given by :

Business Hour	Ganga	Kaveri	Narmada
Hour 1	11 + 12	19	19+12
Hour 2	11+ 10	12+ 10	11 + 10
Hour 3	19+ 10	19+ 11	11 + 12

Additionally, it has been mentioned that no one except Ganga possibly sold Mango smoothies.

Considering Apple Smoothie : (AS), Mango Smoothie : (M.S), Banana Smoothie : (B. S), Mixed Fruit : (M.F), Fruit Salad : (F.S)

Business Hour	Ganga	Kaveri	Narmada
Hour 1	AS + F.S/M.S	M.F	M.F + F.S
Hour 2	AS + B.S	F.S + B.S	AS + B.S
Hour 3	M.F+ B.S	M.F + AS	AS + F.S

In the last one hour everyone had one unit each of two different materials as leftovers:

Each leftover item is sold at Re1 less than their cost price. Hence

One unit of milk, mango, and apple will cost ₹4, ₹2, ₹1,

The three possible combinations for the last hour in which raw materials can be sold is :

One unit of milk+ One unit of mango = Rs 4+2 = Rs 6.

One unit of milk+ One unit of apple = Rs 4+1 = Rs 5.

One unit of mango+ One unit of apple = Rs 2+1 = Rs 3.

In the last one hour :

Total sale of Ganga is Rs 30. This is the total cost earned by considering selling the finished items and raw materials :

Hence the possible cases are :

(24 + 6) or (25 + 5) or (27 + 3). It is not possible to earn 25 and 27 by selling finished products.

Hence she earns Rs 24 which is possible by selling one (M.S + F.S) and one raw unit of milk and mango each.

Total sale of Kaveri is Rs 27. This is the total cost earned by considering selling the finished items and raw materials :

Hence the possible cases are :

(21 + 6) or (22+5) or (24+3).

Since Kaveri cannot sell mango smoothies the possible cases are :

(21+6) Selling one apple smoothie and one banana smoothie and one raw unit of Milk and Mango.

(22+5) Selling one Banana smoothie and Fruit Salad, one unit of raw milk, and one unit of Raw Apple.

Total sale for Narmada is Rs 22. The possible cases are :

(16+6), (17+5), (19+3).

Among these, the only possible case is 19 + 3. Selling one unit Mixed Fruit, and one raw unit of Mango and Apple.

Case 1 :

Business Hour	Ganga	Kaveri	Narmada
Hour 1	A.S + F.S/M.S	M.F	M.F + F.S
Hour 2	A.S + B.S	F.S + B.S	A.S + B.S
Hour 3	M.F+ B.S	M.F + A.S	A.S + F.S
Final Hour	M.S+F.S+Milk+Mango	A.S+B.S+Milk+Mango	M.F+Mango+ Apple

case 2 :

Business Hour	Ganga	Kaveri	Narmada
Hour 1	A.S + F.S/M.S	M.F	M.F + F.S
Hour 2	A.S + B.S	F.S + B.S	A.S + B.S
Hour 3	M.F+ B.S	M.F + A.S	A.S + F.S
Final Hour	M.S+F.S+Milk+Mango	F.S+B.S+Milk+Apple	M.F+Mango+ Apple

3 units of leftover mangoes were available in case 1 and in here a total of 6 apple shakes were sold.

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5.5

Using condition 1 :

Ravi had a total of 21 points in timeliness. 3 of the four customers among Atal, Bihari, Chirag, and Deepak gave him the same ratings. Atal gave the highest rating in timeliness in comparison with Bihari and Chirag. Hence he must have given the distinct rating.

Using condition 2 :

The possibilities are A - 9, B - 4, C - 4, D - 4.

A - 6, B - 5, C - 5, D - 5.

Ravi received a total of 29 points in the Packaging and for this, the possibility of the four scores are (5, 7, 8, 9) awarded by the four customers.

In Hygiene the sum of the ratings awarded was 26. This could possibly be awarded by considering the following cases :

A-(4,6,7,9) , B-(5,6,7,8), C-(4,5,8,9).

Using condition 4: Bihari gave the highest rating in packaging and thus Bihari must have given a 9 rating in the packaging.

Chirag gave the highest rating in Hygiene. In condition 3 it was mentioned that Chirag gave the same points for packaging and hygiene. Since 9 was rated by Bihari packaging it cannot be awarded by Chirag in packaging and hygiene. Since Chirag was awarded the highest in Hygiene. He must award 8 points in Hygiene and Packaging.

Hence of the three possibilities among A, B, and C for Hygiene only B is the possible case with 8 as the maximum score.

In condition 5 it was mentioned that everyone awarded Ravi between 5 and 7 in Behaviour. Unique maximum and minimum ratings in this parameter were given by Atal and Deepak respectively.

Hence Atal must have awarded 7, Deepak 6, Bihari, and Chirag 6 each in Behaviour.

The two possible cases are :

Case 1 :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7			9
Bihari	6	9		4
Chirag	6	8	8	4
Deepak	5			4

Case 2 :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7			6
Bihari	6	9		5
Chirag	6	8	8	5
Deepak	5			5

The ratings awarded by Atal and Deepak in Packaging are among 5 and 7.

The ratings awarded by Atal, Bihari, Deepak are among 5,6, and 7.

Atal individual ranking in Packaging and Hygiene are the same. The same is true for Deepak.

Since Atal and Deepak can give the ranking among 3 and 4 in Packaging as Bihari is first and Chirag is second in this parameter.

They can rank 3 or 4 in the Hygiene parameter also. Hence Bihari must rate 7 points in Hygiene.

In both the possibilities Bihari and Chirag award a total of 26 points. Hence he wins 40 because the total ratings are greater than 25 received from Bihari and Chirag.

Since he gets a total of 120 in bonuses and tips. He must have 80 from Atal and Deepak.

This is possible if he gets a tip of 30 and 50 from them respectively.

In case 1 irrespective of Atal standing at rank 3 or rank 4 in Hygiene and Packaging Atal total rating is greater than 25 which implies Ravi gets a tip from Atal but this is not a possible case because Ravi needs a total of Rs 80 from Atal and Deepak. From Atal if he gets Rs 20 as a bonus he cannot get a total of Rs 120 and hence this case fails.

Hence case 1 fails.

In case 2 there are two possibilities :

Atal ranking 3 in both the parameters and Deepak 4th. Atal ranking 4th in both the parameters and Deepak 3rd

In the case where Atal ranks 3rd in Packaging and Hygiene the total score is 26 and is not a feasible case.

Case - 2A

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	7	6	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	5	5	5

Case - 2B :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	5	5	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	7	6	5

Case - 2A fails because Atal's total rating is greater than 25 which should not be the case.

The minimum rating awarded is 5.

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6.A

Using condition 1 :

Ravi had a total of 21 points in timeliness. 3 of the four customers among Atal, Bihari, Chirag, and Deepak gave him the same ratings. Atal gave the highest rating in timeliness in comparison with Bihari and Chirag. Hence he must have given the distinct rating.

Using condition 2 :

The possibilities are A - 9, B - 4, C - 4, D - 4.

A - 6, B - 5, C - 5, D - 5.

Ravi received a total of 29 points in the Packaging and for this, the possibility of the four scores are (5, 7, 8, 9) awarded by the four customers.

In Hygiene the sum of the ratings awarded was 26. This could possibly be awarded by considering the following cases :

A-(4,6,7,9) , B-(5,6,7,8), C-(4,5,8,9).

Using condition 4: Bihari gave the highest rating in packaging and thus Bihari must have given a 9 rating in the packaging.

Chirag gave the highest rating in Hygiene. In condition 3 it was mentioned that Chirag gave the same points for packaging and hygiene. Since 9 was rated by Bihari packaging it cannot be awarded by Chirag in packaging and hygiene. Since Chirag was awarded the highest in Hygiene. He must award 8 points in Hygiene and Packaging.

Hence of the three possibilities among A, B, and C for Hygiene only B is the possible case with 8 as the maximum score.

In condition 5 it was mentioned that everyone awarded Ravi between 5 and 7 in Behaviour. Unique maximum and minimum ratings in this parameter were given by Atal and Deepak respectively.

Hence Atal must have awarded 7, Deepak 6, Bihari, and Chirag 6 each in Behaviour.

The two possible cases are :

Case 1 :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7			9
Bihari	6	9		4
Chirag	6	8	8	4
Deepak	5			4

Case 2 :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7			6
Bihari	6	9		5
Chirag	6	8	8	5
Deepak	5			5

The ratings awarded by Atal and Deepak in Packaging are among 5 and 7.

The ratings awarded by Atal, Bihari, Deepak are among 5,6, and 7.

Atal individual ranking in Packaging and Hygiene are the same. The same is true for Deepak.

Since Atal and Deepak can give the ranking among 3 and 4 in Packaging as Bihari is first and Chirag is second in this parameter.

They can rank 3 or 4 in the Hygiene parameter also. Hence Bihari must rate 7 points in Hygiene.

In both the possibilities Bihari and Chirag award a total of 26 points. Hence he wins 40 because the total ratings are greater than 25 received from Bihari and Chirag.

Since he gets a total of 120 in bonuses and tips. He must have 80 from Atal and Deepak.

This is possible if he gets a tip of 30 and 50 from them respectively.

In case 1 irrespective of Atal standing at rank 3 or rank 4 in Hygiene and Packaging Atal total rating is greater than 25 which implies Ravi gets a tip from Atal but this is not a possible case because Ravi needs a total of Rs 80 from Atal and Deepak. From Atal if he gets Rs 20 as a bonus he cannot get a total of Rs 120 and hence this case fails.

Hence case 1 fails.

In case 2 there are two possibilities :

Atal ranking 3 in both the parameters and Deepak 4th. Atal ranking 4th in both the parameters and Deepak 3rd

In the case where Atal ranks 3rd in Packaging and Hygiene the total score is 26 and is not a feasible case.

Case - 2A

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	7	6	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	5	5	5

Case - 2B :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	5	5	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	7	6	5

Case - 2A fails because Atal's total rating is greater than 25 which should not be the case.

Deepak gives a rating of 7 in Packaging.

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7.6

Using condition 1 :

Ravi had a total of 21 points in timeliness. 3 of the four customers among Atal, Bihari, Chirag, and Deepak gave him the same ratings. Atal gave the highest rating in timeliness in comparison with Bihari and Chirag. Hence he must have given the distinct rating.

Using condition 2 :

The possibilities are A - 9, B - 4, C - 4, D - 4.

A - 6, B - 5, C - 5, D - 5.

Ravi received a total of 29 points in the Packaging and for this, the possibility of the four scores are (5, 7, 8, 9) awarded by the four customers.

In Hygiene the sum of the ratings awarded was 26. This could possibly be awarded by considering the following cases :

A-(4,6,7,9) , B-(5,6,7,8), C-(4,5,8,9).

Using condition 4: Bihari gave the highest rating in packaging and thus Bihari must have given a 9 rating in the packaging.

Chirag gave the highest rating in Hygiene. In condition 3 it was mentioned that Chirag gave the same points for packaging and hygiene. Since 9 was rated by Bihari packaging it cannot be awarded by Chirag in packaging and hygiene. Since Chirag was awarded the highest in Hygiene. He must award 8 points in Hygiene and Packaging.

Hence of the three possibilities among A, B, and C for Hygiene only B is the possible case with 8 as the maximum score.

In condition 5 it was mentioned that everyone awarded Ravi between 5 and 7 in Behaviour. Unique maximum and minimum ratings in this parameter were given by Atal and Deepak respectively.

Hence Atal must have awarded 7, Deepak 6, Bihari, and Chirag 6 each in Behaviour.

The two possible cases are :

Case 1 :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7			9
Bihari	6	9		4
Chirag	6	8	8	4
Deepak	5			4

Case 2 :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7			6
Bihari	6	9		5
Chirag	6	8	8	5
Deepak	5			5

The ratings awarded by Atal and Deepak in Packaging are among 5 and 7.

The ratings awarded by Atal, Bihari, Deepak are among 5,6, and 7.

Atal individual ranking in Packaging and Hygiene are the same. The same is true for Deepak.

Since Atal and Deepak can give the ranking among 3 and 4 in Packaging as Bihari is first and Chirag is second in this parameter.

They can rank 3 or 4 in the Hygiene parameter also. Hence Bihari must rate 7 points in Hygiene.

In both the possibilities Bihari and Chirag award a total of 26 points. Hence he wins 40 because the total ratings are greater than 25 received from Bihari and Chirag.

Since he gets a total of 120 in bonuses and tips. He must have 80 from Atal and Deepak.

This is possible if he gets a tip of 30 and 50 from them respectively.

In case 1 irrespective of Atal standing at rank 3 or rank 4 in Hygiene and Packaging Atal total rating is greater than 25 which implies Ravi gets a tip from Atal but this is not a possible case because Ravi needs a total of Rs 80 from Atal and Deepak. From Atal if he gets Rs 20 as a bonus he cannot get a total of Rs 120 and hence this case fails.

Hence case 1 fails.

In case 2 there are two possibilities :

Atal ranking 3 in both the parameters and Deepak 4th. Atal ranking 4th in both the parameters and Deepak 3rd

In the case where Atal ranks 3rd in Packaging and Hygiene the total score is 26 and is not a feasible case.

Case - 2A

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	7	6	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	5	5	5

Case - 2B :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	5	5	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	7	6	5

Case - 2A fails because Atal's total rating is greater than 25 which should not be the case.

Atal has given a rating of 6 in timeliness

▶ VIDEO SOLUTION

8. B

Using condition 1 :

Ravi had a total of 21 points in timeliness. 3 of the four customers among Atal, Bihari, Chirag, and Deepak gave him the same ratings. Atal gave the highest rating in timeliness in comparison with Bihari and Chirag. Hence he must have given the distinct rating.

Using condition 2 :

The possibilities are A - 9, B - 4, C - 4, D - 4.

A - 6, B - 5, C - 5, D - 5.

Ravi received a total of 29 points in the Packaging and for this, the possibility of the four scores are (5, 7, 8, 9) awarded by the four customers.

In Hygiene the sum of the ratings awarded was 26. This could possibly be awarded by considering the following cases :

A-(4,6,7,9) , B-(5,6,7,8), C-(4,5,8,9).

Using condition 4: Bihari gave the highest rating in packaging and thus Bihari must have given a 9 rating in the packaging.

Chirag gave the highest rating in Hygiene. In condition 3 it was mentioned that Chirag gave the same points for packaging and hygiene. Since 9 was rated by Bihari packaging it cannot be awarded by Chirag in packaging and hygiene. Since Chirag was awarded the highest in Hygiene. He must award 8 points in Hygiene and Packaging.

Hence of the three possibilities among A, B, and C for Hygiene only B is the possible case with 8 as the maximum score.

In condition 5 it was mentioned that everyone awarded Ravi between 5 and 7 in Behaviour. Unique maximum and minimum ratings in this parameter were given by Atal and Deepak respectively.

Hence Atal must have awarded 7, Deepak 6, Bihari, and Chirag 6 each in Behaviour.

The two possible cases are :

Case 1 :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7			9
Bihari	6	9		4
Chirag	6	8	8	4
Deepak	5			4

Case 2 :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7			6
Bihari	6	9		5
Chirag	6	8	8	5
Deepak	5			5

The ratings awarded by Atal and Deepak in Packaging are among 5 and 7.

The ratings awarded by Atal, Bihari, Deepak are among 5,6, and 7.

Atal individual ranking in Packaging and Hygiene are the same. The same is true for Deepak.

Since Atal and Deepak can give the ranking among 3 and 4 in Packaging as Bihari is first and Chirag is second in this parameter.

They can rank 3 or 4 in the Hygiene parameter also. Hence Bihari must rate 7 points in Hygiene.

In both the possibilities Bihari and Chirag award a total of 26 points. Hence he wins 40 because the total ratings are greater than 25 received from Bihari and Chirag.

Since he gets a total of 120 in bonuses and tips. He must have 80 from Atal and Deepak.

This is possible if he gets a tip of 30 and 50 from them respectively.

In case 1 irrespective of Atal standing at rank 3 or rank 4 in Hygiene and Packaging Atal total rating is greater than 25 which implies Ravi gets a tip from Atal but this is not a possible case because Ravi needs a total of Rs 80 from Atal and Deepak. From Atal if he gets Rs 20 as a bonus he cannot get a total of Rs 120 and hence this case fails.

Hence case 1 fails.

In case 2 there are two possibilities :

Atal ranking 3 in both the parameters and Deepak 4th. Atal ranking 4th in both the parameters and Deepak 3rd

In the case where Atal ranks 3rd in Packaging and Hygiene the total score is 26 and is not a feasible case.

Case - 2A

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	7	6	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	5	5	5

Case - 2B :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	5	5	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	7	6	5

Case - 2A fails because Atal's total rating is greater than 25 which should not be the case.

Atal has given the maximum rating in Behaviour.

 VIDEO SOLUTION

9. B

Using condition 1 :

Ravi had a total of 21 points in timeliness. 3 of the four customers among Atal, Bihari, Chirag, and Deepak gave him the same ratings. Atal gave the highest rating in timeliness in comparison with Bihari and Chirag. Hence he must have given the distinct rating.

Using condition 2 :

The possibilities are A - 9, B - 4, C - 4, D - 4.

A - 6, B - 5, C - 5, D - 5.

Ravi received a total of 29 points in the Packaging and for this, the possibility of the four scores are (5, 7, 8, 9) awarded by the four customers.

In Hygiene the sum of the ratings awarded was 26. This could possibly be awarded by considering the following cases :

A-(4,6,7,9) , B-(5,6,7,8), C-(4,5,8,9).

Using condition 4: Bihari gave the highest rating in packaging and thus Bihari must have given a 9 rating in the packaging.

Chirag gave the highest rating in Hygiene. In condition 3 it was mentioned that Chirag gave the same points for packaging and hygiene. Since 9 was rated by Bihari packaging it cannot be awarded by Chirag in packaging and hygiene. Since Chirag was awarded the highest in Hygiene. He must award 8 points in Hygiene and Packaging.

Hence of the three possibilities among A, B, and C for Hygiene only B is the possible case with 8 as the maximum score.

In condition 5 it was mentioned that everyone awarded Ravi between 5 and 7 in Behaviour. Unique maximum and minimum ratings in this parameter were given by Atal and Deepak respectively.

Hence Atal must have awarded 7, Deepak 6, Bihari, and Chirag 6 each in Behaviour.

The two possible cases are :

Case 1 :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7			9
Bihari	6	9		4
Chirag	6	8	8	4
Deepak	5			4

Case 2 :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7			6
Bihari	6	9		5
Chirag	6	8	8	5
Deepak	5			5

The ratings awarded by Atal and Deepak in Packaging are among 5 and 7.

The ratings awarded by Atal, Bihari, Deepak are among 5,6, and 7.

Atal individual ranking in Packaging and Hygiene are the same. The same is true for Deepak.

Since Atal and Deepak can give the ranking among 3 and 4 in Packaging as Bihari is first and Chirag is second in this parameter.

They can rank 3 or 4 in the Hygiene parameter also. Hence Bihari must rate 7 points in Hygiene.

In both the possibilities Bihari and Chirag award a total of 26 points. Hence he wins 40 because the total ratings are greater than 25 received from Bihari and Chirag.

Since he gets a total of 120 in bonuses and tips. He must have 80 from Atal and Deepak.

This is possible if he gets a tip of 30 and 50 from them respectively.

In case 1 irrespective of Atal standing at rank 3 or rank 4 in Hygiene and Packaging Atal total rating is greater than 25 which implies Ravi gets a tip from Atal but this is not a possible case because Ravi needs a total of Rs 80 from Atal and Deepak. From Atal if he gets Rs 20 as a bonus he cannot get a total of Rs 120 and hence this case fails.

Hence case 1 fails.

In case 2 there are two possibilities :

Atal ranking 3 in both the parameters and Deepak 4th. Atal ranking 4th in both the parameters and Deepak 3rd

In the case where Atal ranks 3rd in Packaging and Hygiene the total score is 26 and is not a feasible case.

Case - 2A

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	7	6	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	5	5	5

Case - 2B :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	5	5	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	7	6	5

Case - 2A fails because Atal's total rating is greater than 25 which should not be the case.

Among Atal and Deepak, one of them gives a tip of 30 and the other gives a tip of 50. Hence 30 or 50 any case is possible

VIDEO SOLUTION

10. C

Using condition 1 :

Ravi had a total of 21 points in timeliness. 3 of the four customers among Atal, Bihari, Chirag, and Deepak gave him the same ratings. Atal gave the highest rating in timeliness in comparison with Bihari and Chirag. Hence he must have given the distinct rating.

Using condition 2 :

The possibilities are A - 9, B - 4, C - 4, D - 4.

A - 6, B - 5, C - 5, D - 5.

Ravi received a total of 29 points in the Packaging and for this, the possibility of the four scores are (5, 7, 8, 9) awarded by the four customers.

In Hygiene the sum of the ratings awarded was 26. This could possibly be awarded by considering the following cases :

A-(4,6,7,9) , B-(5,6,7,8), C-(4,5,8,9).

Using condition 4: Bihari gave the highest rating in packaging and thus Bihari must have given a 9 rating in the packaging.

Chirag gave the highest rating in Hygiene. In condition 3 it was mentioned that Chirag gave the same points for packaging and hygiene. Since 9 was rated by Bihari packaging it cannot be awarded by Chirag in packaging and hygiene. Since Chirag was awarded the highest in Hygiene. He must award 8 points in Hygiene and Packaging.

Hence of the three possibilities among A, B, and C for Hygiene only B is the possible case with 8 as the maximum score.

In condition 5 it was mentioned that everyone awarded Ravi between 5 and 7 in Behaviour. Unique maximum and minimum ratings in this parameter were given by Atal and Deepak respectively.

Hence Atal must have awarded 7, Deepak 6, Bihari, and Chirag 6 each in Behaviour.

Case 1 :

Case 2 :

Case - 2A

Case - 2B :

	Behaviour	Packaging	Hygiene	Timeliness
Atal	7	5	5	6
Bihari	6	9	7	5
Chirag	6	8	8	5
Deepak	5	7	6	5

Case - 2A fails because Atal's total rating is greater than 25 which should not be the case.

Bihari and Chirag has given the highest ratings

[VIDEO SOLUTION](#)



11. C

Based on the given conditions, Damodaran misses out on the bonus if he gets a rating of 1 in any of the five parameters. He additionally needs to obtain a rating of 5 in at least one of the parameters. Thus, the maximum value of the range of ratings that he can acquire would be $1+3(\text{given})+5+5+5$. However, based on condition 2, he can have similar ratings in only two of the parameters. Thus, the maximum value of the final rating would be $(1+3+5+5+4) / 5 = 18/5 = 3.6$. Hence, Option C is the correct answer.

[VIDEO SOLUTION](#)

12. D

Since Eman got a bonus, he must have obtained a rating of 5 in at least one of the parameters. To minimize his final rating we need to consider the following range of values: $5(\text{mandatory})+2(\text{given})+2+3+3 = 15$. The least value of his final rating is, therefore, $15/5 = 3$. Hence, Option D is the correct answer.

[VIDEO SOLUTION](#)

13. D

Our objective here is to minimize the final ratings in order to find the minimum value of the monthly payment. We cannot have a rating of 1 in any of the parameters since all the drives got the bonus and we need to have at least one parameter with a rating of 5. With this understanding, we obtain the following:

Arun:

Final rating = $(5+4+2+2+3)/5 = 16/5 = 3.2$; Fixed payment = Rs.1000

Variable payment = $3.2 * \text{Rs. } 250 = \text{Rs. } 800$; Total = $\text{Rs.}(1000+800) = \text{Rs. } 1800$

Barun:

Final rating = $(5+3+2+2+3)/5 = 15/5 = 3$; Fixed payment = Rs.1200

Variable payment = $3 * \text{Rs. } 200 = \text{Rs. } 600$; Total = $\text{Rs.}(1200+600) = \text{Rs. } 1800$

Chandan: {rating of 5 in exactly two parameters based on condition 1}

Final rating = $(5+5+2+2+3)/5 = 17/5 = 3.4$; Fixed payment = Rs.1400

Variable payment = $3.4 * \text{Rs. } 100 = \text{Rs. } 340$; Total = $\text{Rs.}(1400+340) = \text{Rs. } 1740$

Damodaran:

Final rating = $(5+3+2+2+3)/5 = 15/5 = 3$; Fixed payment = Rs.1300

Variable payment = $3 * \text{Rs. } 150 = \text{Rs. } 450$; Total = $\text{Rs.}(1300+450) = \text{Rs. } 1750$

Eman:

Final rating = $(5+3+2+2+3)/5 = 15/5 = 3$; Fixed payment = Rs.1100

Variable payment = $3 * \text{Rs. } 200 = \text{Rs. } 600$; Total = Rs.(1100+600) = Rs. **1700**

Hence, we observe that the minimum value of the monthly payment is Rs. 1700. Option D is the correct answer.

 VIDEO SOLUTION

14. A

Our objective here is to maximize the final ratings in order to find the maximum possible value of the monthly payment. We cannot have a rating of 1 in any of the parameters since all the drives got the bonus and we need to have at least one parameter with a rating of 5. With this understanding, we obtain the following:

Arun: {can have only one rating of 5 based on condition 1}

Final rating = $(5+4+4+3+3)/5 = 19/5 = 3.8$; Fixed payment = Rs.1000

Variable payment = $3.8 * \text{Rs. } 250 = \text{Rs. } 950$; Total = Rs.(1000+950) = Rs. **1950**

Barun: {can have only one rating of 5 based on condition 1}

Final rating = $(5+4+4+3+3)/5 = 19/5 = 3.8$; Fixed payment = Rs.1200

Variable payment = $3.8 * \text{Rs. } 200 = \text{Rs. } 760$; Total = Rs.(1200+760) = Rs. **1960**

Chandan:

Final rating = $(5+5+2+4+4)/5 = 20/5 = 4$; Fixed payment = Rs.1400

Variable payment = $4 * \text{Rs. } 100 = \text{Rs. } 400$; Total = Rs.(1400+400) = Rs. **1800**

Damodaran:

Final rating = $(5+5+4+4+3)/5 = 21/5 = 4.2$; Fixed payment = Rs.1300

Variable payment = $4.2 * \text{Rs. } 150 = \text{Rs. } 630$; Total = Rs.(1300+630) = Rs. **1930**

Eman:

Final rating = $(5+5+2+4+4)/5 = 20/5 = 4$; Fixed payment = Rs.1100

Variable payment = $4 * \text{Rs. } 200 = \text{Rs. } 800$; Total = Rs.(1100+800) = Rs. **1900**

Hence, we observe that the maximum possible value of the monthly payment is Rs. 1960. Option A is the correct answer.

 VIDEO SOLUTION

15. D

Given that each of the bottles contains a volume of 50 ml each.

If 5 ml from bottle A which contains only P is mixed with 5 ml of bottle B and in the resultant mixture the presence of I was detected.

In order to detect the presence of I in this, there must be at least 10% impurity in the 10 ml which is equivalent to 1 ml. This must be from bottle B.

Hence 5 ml of solution from B must contain at least 1ml of impurity and since bottle B is of a total volume of 50 ml. This must contain at least 10 ml of impurity.

 VIDEO SOLUTION



CAT Daily Targets

Step by step, heading towards 99+ percentile



16.D

The bottles could possibly be :

Case - 1 Pure, Impure, Impure, Impure.

Case-2, Pure , Pure, Impure, Impure.

Since the concentration in the impure bottle is 85 percent.

In case 1 when equal volumes from all the bottles are considered and mixed. The test result detects the impurity..

Since the overall concentration of impurity is greater than 10 percent.

Considering 10 ml from all four bottles.

The impure concentration is $4.5\text{ml}/40\text{ml}$ which is greater than 10. ($15\text{ml} \times 3 = 4.5\text{ml}$) (Impurity is detected)

For case 2 when all four bottles are considered. The case here has 2 pure and 2 impure bottles.

When equal volumes from all four bottles are mixed. The resultant concentration of impurity when 10 ml from each of the four solutions is considered :

The impure concentration is $3\text{ml}/40\text{ml}$ which is less than 10 percent.. ($1.5\text{ml} \times 2 = 3\text{ml}$). (Impurity is not detected.)

Hence in one possibility the impurity is detected and not detected in the other case. A single test is enough based on the results of which the number of pure and the number of impure bottles can be identified.

 VIDEO SOLUTION

17.2

The percentage concentration of the impure solution is 80 percent.

When equal volumes of all four solutions are mixed.

Considering 10 ml of each we have impurity to be $2\text{ml}/40\text{ml}$. The impurity concentration is less than 10 percent and hence cannot be recognized.

Similarly when equal volumes of one impure and 2 pure solutions are mixed.

The impurity in the solution is $2\text{ml}/30\text{ml}$ which is less than 10 percent and hence cannot be recognized.

Hence for detecting the impure solution we must use equal volumes of 2 solutions at a time.

Considering the three pure solutions to be P and the impure solution to be I.

P, P, P, I.

Considering equal volumes of solution from the bottle one bottle of P, and I. Testing this would recognize the impurity.

After this consider one bottle among the other 2 P bottles which are left and test this with one among the previously tested P, I.

If the one considered is I it will detect the impurity and confirms the bottle to be I.

If the one considered is P it will fail to detect the impurity and hence the other bottle will be I.

Hence a minimum of two tests are required to identify the bottle with the impurity.

 VIDEO SOLUTION

18.1

The bottles contain either P(pure) or I(impure). The possible cases here are :

1- (P, P, P, P), 2-(P,P,P,I), 3-(P,P,I,I), 4-(P,I,I,I), 5-(I,I,I,I).

In the first case if all the four solutions are pure then taking equal volumes of all the four bottles will get the result to dispatch or not to dispatch.

In the second case if 3 bottles are pure and one impure taking equal volumes of all four bottles and testings will confirm the impurity and hence cannot be dispatched.

In the third case if 2 bottles are pure and two are impure taking equal volumes of all four bottles and testing will confirm the impurity and hence cannot be dispatched.

In the fourth case when only one bottle is pure taking equal volumes of all four bottles will confirm the impurity and hence cannot be dispatched.

In the fifth case if all four bottles are impure taking equal volumes of the four bottles will confirm the impurity and hence cannot be dispatched.

In all the cases a single test is enough to determine if the lot is to be dispatched or not.

 VIDEO SOLUTION

19.2

It is given that the most expensive item is a diamond ring of type a and there is exactly one of these. Since the item b should be at least twice. The minimum number of items will be obtained when $a=1$ and $b=99$, which means there are only two different types of items.

 VIDEO SOLUTION

20.6

It is given that the most expensive item is a diamond ring of type a and there is exactly one of these. Since the number of items of type b should be at least twice of that of a and the number of items of type c should be at least twice of that of b and so on. So the maximum number of different types of items of a, b and c will be obtained when $a=1$, $b=2$, $c=4$, $d=8$, $e=16$, $f=69$. Hence the maximum number of different types of items will be 6.

If the number of items is 7, then the minimum number of prizes should be $1+2+4+8+16+32+64=127$ which is more than 100.

Hence 6 is the answer.

 VIDEO SOLUTION



21. C

Option A: There are exactly 75 items of type e.

$a=1, b=2, c=4, d=8, e=85$. Here the maximum value of $e=85$. Hence it can take the value 75.

An example of such case is $a=1, b=2, c=4, d=18, e=75$

Option B: There are exactly 30 items of type b.

$a=1$ $b=30$ and $c=69$. Hence this case is also possible.

Option C: There are exactly 45 items of type c.

Since the value of d should be at least 90, it means that d is not present because $45+90$ will be more than 100 (maximum number of items). Only a, b and c are present.

The maximum value of $b=22$ and $a=1$, but $45+22+1=68$, which is less than 100. So this case is not possible.

Option D: There are exactly 60 items of type d.

$d=60, c=30, b=9$ and $a=1$. $a+b+c+d=100$. Hence this case is possible.

C is the answer.

 VIDEO SOLUTION

22. **A**

The total number of items from 1 to 100, which are of same type as in box 45 = $31+1+43=75$

Now to maximize the number of items, $a=1, b=2, c=4, d=18$ and $e=75$ (given)

There can be maximum 5 types of items.

If we consider number of items to be 6, then minimum number of items of 5th type will be 16,
 $1+2+4+8+16+75=106$ which is more than 100.

 VIDEO SOLUTION

23. **33**

Let us assume, A is the total number of AC's sold

=> From the information that the total number of ACs sold in the city, 25% were of Window variant => Window AC's = $A/4$ and Split AC's = $3A/4$

Now, let us assume B is the total number of inverter ACs

=> From the information that among the Inverter ACs sold, 20% were of Window variant. => Window Inverter AC's = $B/5$ and Window Non-Inverter AC's = $4B/5$

Total (A)			
Split ($3A/4$)		Window ($A/4$)	
Inv ($4B/5$)	Non-Inv	Inv ($B/5$)	Non-Inv

From - Condition-3

=> $A/4 - B/5 = 6$ and $4B/5 = 36$ => $B = 46$ and $A = 60$.

Total = 60			
Split = 45		Window = 15	
Inv = 36	Non-Inv = 9	Inv = 9	Non-Inv = 6

Now, from condition-6

a) D1 & D4 sold "0" window Non-inverter ACs => D2 & D3 sold 6 window non-inverter ACs, it is given that D2 sold twice as many as D3 => D2 sold 4 and D3 sold 2 ACs of this type.

From condition-2

b) Let us assume, D1 sold "x" window inverter ACs => Number of split inverter ACs sold is $13-x$

From condition-4

c) Number of split ACs sold by D1 will be "2x"

From condition-5

d) Let us assume 'y' is the number of window ACs sold by D3 & D4 => D2 sold 3y ACs of this type.

From condition-7

e) Let us assume 'z' is the number of split inverter ACs sold by D3 and D4 => D2 sold 2z ACs of this type.

Let us use a, b, c, d, and e make a table:

D1 Total =			
Split =		Window = x	
Inv = 13-x	Non-Inv =	Inv = x	Non-Inv = 0

D2 Total =			
Split =		Window = 3y	
Inv = 2z	Non-Inv =	Inv =	Non-Inv = 4

D3 Total =			
Split =		Window = y	
Inv = z	Non-Inv = 3	Inv =	Non-Inv = 2

D4 Total =			
Split =		Window = y	
Inv = z	Non-Inv =	Inv =	Non-Inv = 0

We know that the total number of window ACs is 15

$\Rightarrow x + 3y + y + y = 15 \Rightarrow x + 5y = 15$, also x and y should be greater than or equal to 2 from condition-1

$\Rightarrow x = 5$ and $y = 2$ is the only solution.

Filling this in the table:

D1 Total =			
Split =		Window = 5	
Inv = 8	Non-Inv =	Inv = 5	Non-Inv = 0

D2 Total =			
Split =		Window = 6	
Inv = 2z	Non-Inv =	Inv = 2	Non-Inv = 4

D3 Total =			
Split =		Window = 2	
Inv = z	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total =			
Split =		Window = 2	
Inv = z	Non-Inv =	Inv = 2	Non-Inv = 0

Now, Number of split inverter ACs is 36

$\Rightarrow 8 + 2z + z + z = 36 \Rightarrow 4z = 28 \Rightarrow z = 7$.

Filling this and using (5), the number of split AC's sold by D1 is $2 \times 5 = 10$.

D1 Total = 15			
Split = 10		Window = 5	
Inv = 8	Non-Inv = 2	Inv = 5	Non-Inv = 0

D2 Total =			
Split =		Window = 6	
Inv = 14	Non-Inv =	Inv = 2	Non-Inv = 4

D3 Total = 12			
Split = 10		Window = 2	
Inv = 7	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total =			
Split =		Window = 2	
Inv = 7	Non-Inv =	Inv = 2	Non-Inv = 0

Total number of ACs sold by D2 and D4 = $60 - D1 - D3 = 60 - 15 - 12 = 33$.

▶ VIDEO SOLUTION

24. C

Let us assume, A is the total number of AC's sold

=> From the information that the total number of ACs sold in the city, 25% were of Window variant => Window AC's = $A/4$ and Split AC's = $3A/4$

Now, let us assume B is the total number of inverter ACs

=> From the information that among the Inverter ACs sold, 20% were of Window variant. => Window Inverter AC's = $B/5$ and Window Non-Inverter AC's = $4B/5$

Total (A)			
Split ($3A/4$)		Window ($A/4$)	
Inv ($4B/5$)	Non-Inv	Inv ($B/5$)	Non-Inv

From - Condition-3

=> $A/4 - B/5 = 6$ and $4B/5 = 36$ => $B = 46$ and $A = 60$.

Total = 60			
Split = 45		Window = 15	
Inv = 36	Non-Inv = 9	Inv = 9	Non-Inv = 6

Now, from condition-6

a) D1 & D4 sold "0" window Non-inverter ACs => D2 & D3 sold 6 window non-inverter ACs, it is given that D2 sold twice as many as D3 => D2 sold 4 and D3 sold 2 ACs of this type.

From condition-2

b) Let us assume, D1 sold "x" window inverter ACs => Number of split inverter ACs sold is $13-x$

From condition-4

c) Number of split ACs sold by D1 will be "2x"

From condition-5

d) Let us assume 'y' is the number of window ACs sold by D3 & D4 => D2 sold $3y$ ACs of this type.

From condition-7

e) Let us assume 'z' is the number of split inverter ACs sold by D3 and D4 => D2 sold $2z$ ACs of this type.

Let us use a, b, c, d, and e make a table:

D1 Total =			
Split =		Window = x	
Inv = $13-x$	Non-Inv =	Inv = x	Non-Inv = 0

D2 Total =			
Split =		Window = $3y$	
Inv = $2z$	Non-Inv =	Inv =	Non-Inv = 4

D3 Total =			
Split =		Window = y	
Inv = z	Non-Inv = 3	Inv =	Non-Inv = 2

D4 Total =			
Split =		Window = y	
Inv = z	Non-Inv =	Inv =	Non-Inv = 0

We know that the total number of window ACs is 15

=> $x + 3y + y + y = 15$ => $x + 5y = 15$, also x and y should be greater than or equal to 2 from condition-1

=> $x = 5$ and $y = 2$ is the only solution.

Filling this in the table:

D1 Total =			
Split =		Window = 5	
Inv = 8	Non-Inv =	Inv = 5	Non-Inv = 0

D2 Total =			
Split =		Window = 6	
Inv = 2z	Non-Inv =	Inv = 2	Non-Inv = 4

D3 Total =			
Split =		Window = 2	
Inv = z	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total =			
Split =		Window = 2	
Inv = z	Non-Inv =	Inv = 2	Non-Inv = 0

Now, Number of split inverter ACs is 36

$$\Rightarrow 8 + 2z + z + z = 36 \Rightarrow 4z = 28 \Rightarrow z = 7.$$

Filling this and using (5), the number of split AC's sold by D1 is $2 \times 5 = 10$.

D1 Total = 15			
Split = 10		Window = 5	
Inv = 8	Non-Inv = 2	Inv = 5	Non-Inv = 0

D2 Total =			
Split =		Window = 6	
Inv = 14	Non-Inv =	Inv = 2	Non-Inv = 4

D3 Total = 12			
Split = 10		Window = 2	
Inv = 7	Non-Inv = 3	Inv = 0	Non-Inv = 2

D4 Total =			
Split =		Window = 2	
Inv = 7	Non-Inv =	Inv = 2	Non-Inv = 0

Total = 60			
Split = 45		Window = 15	
Inv = 36	Non-Inv = 9	Inv = 9	Non-Inv = 6

From this table, we see that total number of non-inverter ACs is $9 + 6 = 15$.

Required percentage is 15 out of 60 $\Rightarrow 25\%$.

 VIDEO SOLUTION